



FSO as backhaul and energizer for drone-assisted mobile access networks[☆]

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Abstract

Drone mounted base stations (DBSs) can be flexibly deployed over some areas (such as hotspots and disaster struck areas) to speed up the communication between terrestrial base stations (TBSs) and users. In order to increase the capacity of the backhaul link between a DBS and its TBS, free space optical (FSO) communications is applied as the backhaul solution. However, the DBS and its TBS should be in line-of-sight (LoS). Thus, a new DBS placement method is introduced to maximize the number of served users while guaranteeing the LoS between the DBS and its TBS. In addition, in order to prolong the hovering time of a DBS, we propose to deploy an optical beam to facilitate simultaneous communications and charging. That is, an optical beam transmitted from a TBS to its DBS carries not only data but also energy. The DBS mounted with a solar panel and an FSO receiver receives both the data and energy carried by the optical beam simultaneously. Numerical results demonstrate that using an optical beam to charge and communicate with a DBS can gain 25% extra hovering time of the DBS and achieve high network throughput.

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Keywords: Drone assisted access network; Free space optical communications; Wireless charging

1. Introduction

According to Ericsson Mobility Report [1], the number of mobile broadband subscriptions has been growing at 15% year-on-year, increased by 140 million in the first quarter of 2019. Accordingly, the mobile traffic growth is exploding owing to the rising number of mobile broadband subscriptions and an increasing average data volume per subscription. Mobile traffic is expected to grow by 25% annually between 2018 and 2024. The current mobile network infrastructure is facing a very big challenge to accommodate the exponential growth of mobile traffic [2]. In order to significantly increase the capacity of mobile networks, various technologies, such as ultra-dense small cell deployment, mmWave communications, and massive MIMO, have been incorporated in 5G mobile networks [3]. Drone assisted mobile access networks are one of

the promising technologies to quickly and efficiently increase the network capacity.

In a drone assisted mobile access network, a drone mounted base station (DBS) is deployed over a specific area, serving as a relay node between a remote terrestrial base station (TBS) and users in the area [4–8]. Specifically, as shown in Fig. 1, the requested data streams are transmitted from the remote TBS to the DBS via the wireless backhaul link [9]. Then, the DBS relays the received data streams to the corresponding users via the wireless access links. Since the DBS can be deployed very close to the users, the data rates of users in downloading data streams from the DBS are much higher than those from the remote TBS.

Attributed to the flexible and quick deployment of DBSs, drone assisted mobile access networks have been applied for various application scenarios. First, DBSs can be deployed over random and unpredictable hotspots, in which deploying small cells is neither economical nor practical [10]. For example, a heavy traffic jam area may become a hotspot, where users may require to download and watch various contents when they wait in traffic jams. Thus, deploying DBSs over the heavy traffic jam area may significantly increase the data

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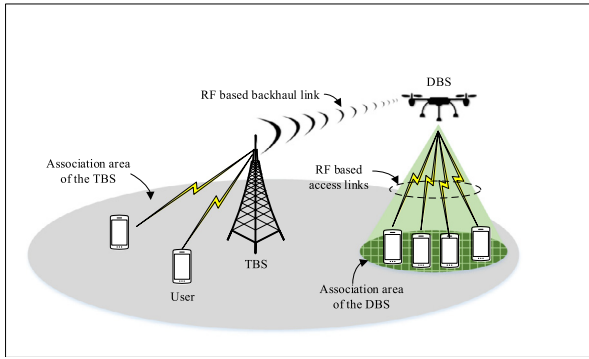


Fig. 1. The drone assisted access network architecture.

rates of users. *Second*, DBSs can be deployed over a disaster struck area, where wireless network infrastructure may be damaged and the disaster struck area is hard to reach from the ground [11]. For example, an earth quake may strike and destroy the network infrastructure in an area, where users are isolated from the outside world. By deploying DBSs over the area, a temporary network can be quickly established to enable victims to contact, for example, the rescue team. *Third*, DBSs can be used to data collection in a wireless sensor network, where many sensor nodes are deployed in a large monitoring area [12]. DBSs can periodically cruise over the area to retrieve sensed data from these sensor nodes to save energy consumption of the sensor nodes.

Drone assisted mobile access networks can facilitate many applications; however, there are several unveiled research challenges. (1) A DBS can be deployed very close to the users to increase the data rates between the DBS and users. However, the wireless backhaul link could become the bottleneck, thus reducing the overall network throughput. How to increase the wireless backhaul link capacity without affecting the wireless access link capacity is a challenging problem. (2) Deploying a DBS in different locations may significantly affect the network performance. That is, deploying the DBS closer to users may reduce the wireless link capacity between the DBS and the remote TBS. Yet, deploying the DBS closer to the remote TBS may reduce the wireless access link capacities between the DBS and the users. It is necessary to find an optimal DBS deployment to maximize the network capacity. (3) A DBS is normally powered by its portable battery, which limits the hovering time of the DBS. For example, DJI PHANTOM 4 can only fly for around 28 min and Yuneec Q500 has the maximum flight time of 25 min. The short battery life of a DBS thus limits its hovering time to assist the TBS in delivering traffic.

In this paper, we investigate FSO based drone assisted access networks, where FSO communications is applied as the backhaul solution to significantly increase the backhaul link capacity. Also, a DBS placement method is introduced to efficiently deploy multiple DBSs in the context of an FSO based drone assisted access network. In addition, we propose to use an optical beam to achieve simultaneous charging and communications with a DBS such that the hovering time and the backhaul link capacity can be increased. Numerical results

show that applying optical charging can extend the hovering time of a DBS over 25%.

The rest of this paper is organized as follows. In Section 2, we introduce the FSO based drone assisted access network architecture and a QoS aware multiple DBS placement method. In Section 3, we introduce the framework of achieving simultaneous charging and communications with a DBS and develop system models to estimate the charging rate and data rate of applying an optical beam to carry energy and data. In Section 4, we provide numerical results to demonstrate that a DBS can gain 25% extra hovering time by applying optical charging. A brief conclusion is drawn in Section 5.

2. Applying FSO as the backhaul solution in drone assisted mobile access networks

As mentioned earlier, a DBS may be deployed far away from its TBS, thus leading to the bottleneck in the backhaul link between the DBS and its TBS. In order to increase the backhaul link capacity, free space optical (FSO) communications is proposed to be the solution. As compared to the traditional RF communications, FSO can achieve higher capacity over longer distance. It has been experimentally demonstrated that an FSO link can offer 1.28 Tbps data rate between the two end points at a distance of 210 m [13]. Although applying FSO can significantly improve the backhaul link capacity, a new challenge arises, i.e., FSO is line-of-sight (LoS) communications, and so the link between a DBS and its TBS should be LoS. Thus, new DBS placement strategies should be designed to guarantee the LoS between a DBS and its associated TBS.

We have investigated the DBS placement problem in the context of a disaster struck area [14]. Specifically, as shown in Fig. 2, all the TBSs in the disaster struck area are damaged, and so a number of DBSs are placed over the disaster struck area to help the users download data from remote TBSs, which are located out of the disaster struck area. That is, the data streams are downloaded from a remote TBS to its DBS via the FSO based backhaul link, and then the DBS relays the received data streams to its associated users via wireless access links.

Given the number of available DBSs, we formulate the problem by jointly optimizing the DBS deployment, user association, and bandwidth allocation such that the number of served users in the disaster struck area is maximized [14]. Here, a user that can be served by a DBS if the achievable data rate of the user (in downloading data from the DBS) can meet its data rate requirement. We have designed the QoS aware drone base Station placement and mobile User association strategy (RESCUE) to efficiently solve the problem [14]. Basically, RESCUE comprises three major steps. (1) DBSs are first iteratively deployed over the area to maximize the their coverage area (i.e., maximize the number of its covered users who have not been covered by the deployed DBSs); (2) given the total amount of available bandwidth for each DBS, each DBS allocates bandwidth to its covered users in order to maximize the number of served users; (3) adjust the altitude of each DBS such that the number of served users for the DBS can be increased. The simulation results show that, as compared to

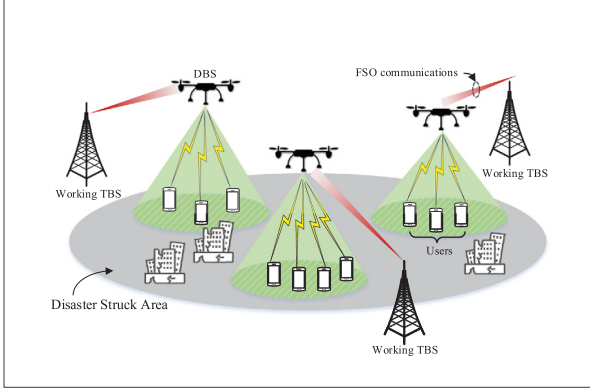


Fig. 2. Applying an FSO based drone assisted network in a disaster struck area.

the existing traffic load aware DBS configuration (TLA)¹ [15] and pathloss aware DBS configuration (PLA)² [16] methods, RESCUE achieves the most number of served users in the disaster struck area and the highest network throughput [14].

3. Simultaneous communications and charging

Applying FSO as backhaul communications can significantly increase the backhaul link capacity, and thus dramatically improve the network throughput. However, how to extend the hovering time of a DBS is still unveiled. We proposed the SoarNet architecture, where the optical beam transmitted from a TBS to its DBS carries not only data but also energy [17]. That is, the DBS can simultaneously receive data and energy transmitted from the TBS, thus prolonging the hovering time of the DBS while providing a high backhaul link capacity. Specifically, as shown in Fig. 3, a DBS is mounted with an optical beam receiver, which comprises an FSO receiver and a solar panel. The FSO receiver receives one part of the optical beam from the TBS and demodulates the data carried by the optical beam. The solar panel receives the rest optical beam and absorbs the energy carried by the optical beam. The absorbed energy is stored in the portable battery of the DBS for further usage. As compared to traditional wireless charging methods, applying an optical beam to charge a battery has a higher charging efficiency. It has been demonstrated that only 3.4% RF energy can be transformed into electrical energy [18]; however, 20% optical energy can be converted into electrical energy [19]. This is because an optical beam used in FSO communications has a very narrow divergence angle, thus making the optical energy more concentrated. The divergence angle of an optical beam in FSO can be as small as 0.02 rad [20].

¹ The basic idea of TLA is to maximize the overall data rate between the DBSs and their users by first allocating bandwidth to the users with lower pathloss to their associated DBSs. However, TLA does not yield the optimal DBS deployment, and thus the locations of the DBSs, derived from RESCUE, will be applied to TLA.

² The basic idea of PLA is to jointly optimize the bandwidth allocation and horizontal locations of DBSs in order to minimize the average pathloss between the user and their associated DBSs. However, the altitudes of the DBSs are fixed and predefined.

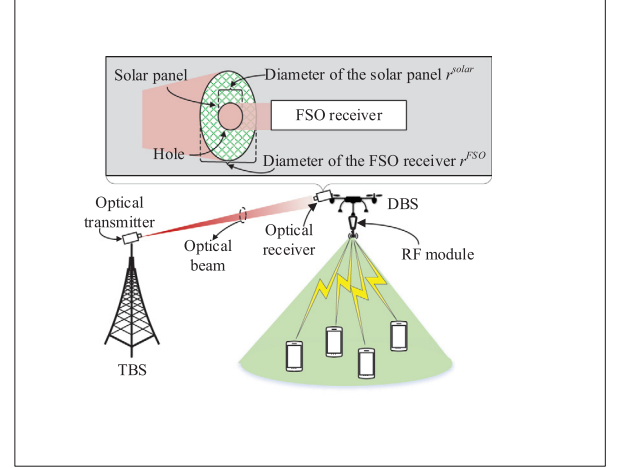


Fig. 3. Simultaneous communications and charging via an optical beam.

3.1. Received optical power

In this section, we will provide a mathematical model to estimate the charging rate of using an optical beam to charge a DBS. Denote P^r as the power of the optical beam received by a DBS. Let P^d and P^c be the received optical power used for communications and charging, respectively. We assume that the size of the optical beam spot is the same as the size of the optical beam receiver in the DBS. Then, we have

$$P^r = P^d + P^c. \quad (1)$$

Here, the optical power received by the optical receiver in the DBS can be estimated by [21]

$$P^r = P^t \cdot \eta^t \cdot 10^{-\frac{L_{poi}}{10}} \cdot 10^{-\frac{L_{atm}}{10}}, \quad (2)$$

where P^t is the transmission power of an optical beam, η^t is the efficiency of the FSO transmitter in the TBS, L_{poi} is the pointing loss³ of the optical beam, and L_{atm} is the scattering loss⁴, which is mainly determined by the distance between the TBS and the DBS as well as the visibility range (i.e., the maximum distance that one object can be clearly discerned) of the environment, i.e.,

$$L_{atm} = 4.34 \cdot \sigma \cdot d, \quad (3)$$

where d is the distance between the TBS and the DBS, and σ is the atmospheric attenuation coefficient in km^{-1} , which is estimated by [23]

$$\sigma = \frac{3.91}{v} \cdot \left(\frac{\lambda}{550} \right)^{-q}, \quad (4)$$

where λ is the wavelength of the optical beam in nm, v is the visibility range in km, and q is the size distribution of the scattering particles. Here, q can be estimated based on the

³ Here, we assume that the Acquisition, Tracking, and Pointing (ATP) system [22] is applied, and thus the pointing loss is not considered in this paper.

⁴ The units of L_{poi} and L_{atm} are in dB.

visibility range v [23], i.e.,

$$q = \begin{cases} 1.6, & v > 50 \text{ km}, \\ 1.3, & 6 \text{ km} < v \leq 50 \text{ km}, \\ 0.16v + 0.34, & 1 \text{ km} < v \leq 6 \text{ km}, \\ v - 0.5, & 0.5 \text{ km} < v \leq 1 \text{ km}, \\ 0, & v \leq 0.5 \text{ km}. \end{cases} \quad (5)$$

Note that the visibility range v is related to the weather conditions. That is, a clear weather implies $v > 50$ km, a hazy weather indicates $1 \text{ km} < v \leq 6 \text{ km}$, and a foggy weather means $v \leq 1 \text{ km}$.

Thus, given the visibility range v of the environment, the distance between the TBS and the DBS d , and the wavelength of the optical beam λ , the optical power received by the optical receiver in the DBS can be estimated based on Eqs. (2), (3), (4), and (5).

We assume that the energy of the optical beam spot is uniformly distributed. Then, as shown in Fig. 3, the values of P^d and P^c depend on the area sizes of the solar panel and the FSO receiver, i.e.,

$$\frac{P^d}{P^c} = \frac{(r^{FSO})^2}{(r^{solar})^2 - (r^{FSO})^2} = \frac{\alpha}{1 - \alpha}, \quad (6)$$

where r^{solar} is the radius of the solar panel, r^{FSO} is the radius of the FSO receiver, and α is the square of the ratio of r^{FSO} to r^{solar} , i.e.,

$$\alpha = \left(\frac{r^{FSO}}{r^{solar}} \right)^2. \quad (7)$$

3.2. Data rate model of an optical beam

The achievable data rate of an FSO link between a DBS and a TBS is estimated by [24]

$$R = \frac{\eta^r \cdot P^d}{E_p \cdot N_b}, \quad (8)$$

where N_b is the sensitivity of the FSO receiver (photons/bit), η^r is the optical efficiency of the FSO receiver in the DBS and E_p is the energy of each photon, i.e., $E_p = h_p c / \lambda$ (here, h_p is the Planck's constant and c is the light speed).

3.3. Charging rate model of an optical beam

The optical power received by the solar panel can be converted into electrical energy stored in the portable battery of the DBS. Let γ be the efficiency of using the received optical power to power the DBS, which equals the efficiency of converting the optical power into the electrical power times the efficiency of charging the portable battery by the converted electrical power. Thus, the rate of charging the portable battery in the DBS (denoted as $\hat{P}^c$) is

$$\hat{P}^c = \gamma \cdot P^c. \quad (9)$$

3.4. Extra hovering time

The optical beam is used to transmit not only data but also energy to a DBS. Thus, it is worth to investigate that how long the hovering time of a DBS can be extended while using an optical beam to charge the DBS. Denote W and P^h as the current residual energy in a DBS's portable battery and the power consumption of the DBS hovering over a specific location, respectively. Let T be the maximum hovering time of the DBS (i.e., the period from now until the portable battery is empty). Then, we have $W + T \cdot \hat{P}^c = T \cdot P^h$, i.e.,

$$T = \frac{W}{P^h - \hat{P}^c}. \quad (10)$$

Thus, the extra hovering time for applying optical charging (ΔT) is

$$\Delta T = \frac{W}{P^h - \hat{P}^c} - \frac{W}{P^h}. \quad (11)$$

4. Numerical results

We assume that the transmission power of the optical beam P^t is 100 W, the sensitivity of the FSO receiver in a DBS (i.e., N_b) is 100 photons/bit, the wavelength of an optical beam (i.e., λ) is 1550 nm, the charging efficiency (i.e., γ) is 15%, the hovering power consumption of a DBS (i.e., P^h) is 50 W [25], and the remaining energy in the DBS's portable battery (i.e., W) is 30 Wh. Thus, the expected hovering time of the DBS without having the optical charging is 30 Wh/50 W = 36 min.

Fig. 4 shows how the extra hovering time varies with respect to the visibility range (i.e., v) and the size of solar panel/FSO receiver (i.e., α), when the distance between the DBS and the TBS is 1 km. When $500 \text{ m} \leq v$, the optical beam does not suffer from severe scattering loss, and so the solar panel in the DBS can receive high optical power to charge its battery, thus significantly prolonging its hovering time. For example, the DBS can have extra 8 min hovering time when $v = 2 \text{ km}$ and $\alpha = 0.5$. That is, the hovering time is increased over 25% as compared to the case that the optical beam is not used to charge the DBS. However, once the visibility range v decreases, the extra hovering time reduces accordingly. When $v < 800 \text{ m}$, the solar panel can barely receive the optical beam and convert it into electrical energy owing to the severe scattering loss. Thus, the DBS gains little extra hovering time. Also, as α reduces, the size of the solar panel reduces, and so the received optical power declines. Accordingly, the extra hovering time of the DBS decreases.

Next, we evaluate how the throughput between a user and its TBS (via a DBS) and the extra hovering time ΔT change by varying the distance between the DBS and the TBS. Assume that a user is placed at a location whose distance to the TBS is 1 km, and the DBS is placed between the TBS and the user. The amount of bandwidth allocated to the user for downloading traffic from the DBS is 5 MHz. Also, the visibility range v and α are set to be 800 m and 0.5, respectively. Fig. 5 shows the throughput of the user in downloading traffic from the TBS and the extra hovering time

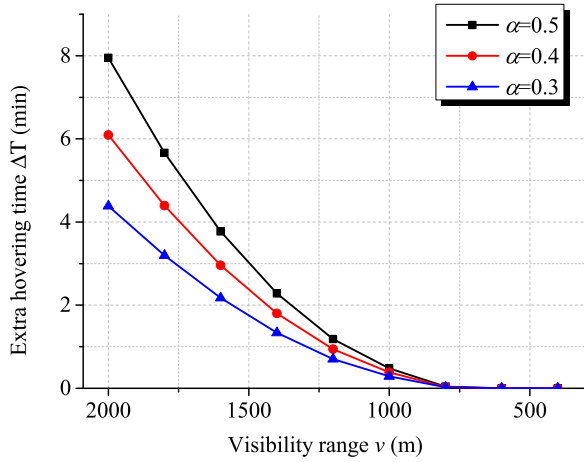


Fig. 4. Extra hovering time over visibility range.

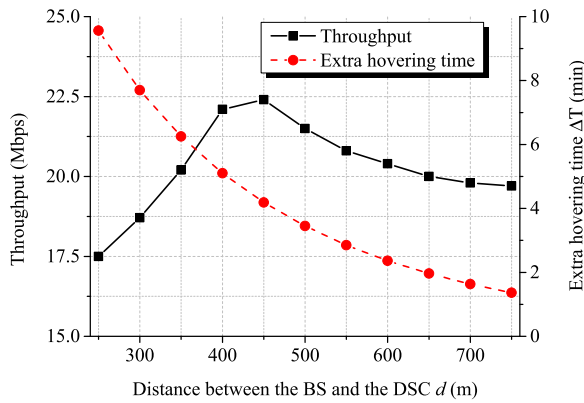


Fig. 5. Trade-off between throughput and extra hovering time.

of the DBS when we move the DBS from the TBS towards the user. As the horizontal distance between the TBS and the DBS increases, the scattering loss increases, and thus the optical power received by the solar panel reduces. As a result, the extra hovering time of the DBS reduces. On the other hand, as the horizontal distance increases, the throughput of the FSO backhaul link (between the TBS and the DBS) reduces, but the throughput of the access link (between the DBS and the user) increases. So, when $d \leq 800$ m, the access link is the bottleneck of the path from the TBS to the user via the DBS. Increasing d improves the throughput of the access link, thus increasing the throughput of the path. When $d \geq 900$ m, the backhaul link becomes the bottleneck of the path. Increasing d reduces the throughput of the backhaul link, and thus declines the throughput of the path.

5. Conclusion

In this paper, we have proposed to use of FSO as a backhaul link between a TBS and a DBS to achieve the simultaneous charging and communications with the DBS. Thus, the data rate and the hovering time of the DBS are significantly increased. Numerical results demonstrate that use of an optical

beam to charge and communicate with a DBS can gain 25% extra hovering time of the DBS and achieve high network throughput.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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